

Willow McPhail

New York, NY | 914-806-1544 | mcphailwillow@gmail.com | git: w-mcphail

SUMMARY

Quantitative researcher and trader combining independent research in factor signals, statistical arbitrage, and walk-forward validation with live-market experience in systematic and discretionary futures trading. Seeking a quant research / trading role where rigorous signal research and disciplined risk management translate directly.

EDUCATION

B.S. Computer Science, Hunter College | 2019 - 2024 | New York, NY

GPA: 3.5/4.0; **SAT:** Math 800; **SAT Math II:** 800

QUANTITATIVE RESEARCH

Crypto Statistical Arbitrage Research (Python, pandas, NumPy, statsmodels, scipy)

- Achieved Sharpe 0.41 net of 10 bps/turnover costs on a daily-rebalanced cross-sectional long/short momentum strategy across a 20-asset crypto universe, walk-forward validated over 2 years of daily bars
- Engineered a 20-signal library spanning cross-sectional momentum, BTC-adjusted excess momentum, time-series momentum, volume-confirmed momentum, and mean reversion; measured signal quality via Information Coefficient (IC) and ICIR across all signals
- Applied a net-Sharpe quality gate to filter qualifying signals; compared six portfolio construction methods (equal weight, rank-normalize, regime-filtered, Sharpe-weighted, IC-weighted, mean-variance)
- Validated with three independent walk-forward frameworks covering signal-window selection, portfolio construction pipeline, and pairs selection; identified regime-dependency as the primary failure mode (strategy profitable in trending markets, negative in choppy/macro-shock regimes)
- Built a full pairs stat-arb pipeline (Engle-Granger cointegration, expanding-OLS hedge ratios, spread z-score entry/exit); diagnosed spurious cointegration in crypto (185/190 pairs flagged at $p < 0.05$) and IS/OOS Sharpe gap (1.6 IS $\rightarrow$ -0.14 OOS walk-forward), demonstrating selection bias identification

PROFESSIONAL EXPERIENCE

Quantitative / Futures Trader, Self-Directed | May 2024 - Present | New York City, NY

- Designed, backtested, and deployed 30+ systematic trading strategies across crypto futures (momentum, mean-reversion, and statistical signals), prioritizing out-of-sample validation and transaction-cost modeling over in-sample fit.
- Traded CME E-mini Nasdaq-100 (NQ) futures discretionarily, documenting repeatable asymmetric setups (avg 2.5 R:R, ~2:1 winner-to-loser) and translating market-microstructure intuition into testable, rules-based hypotheses.
- Maintained disciplined live-capital risk management, holding maximum drawdown to 20% at a 1.7 profit factor across the deployment period of 2 years.

Founder, Effortless Home Maintenance | August 2026 - Present | New York, NY

- Designed, built, and deployed a production full-stack web application solo: TypeScript/React front end, PostgreSQL schema and migrations, user authentication, TLS and secrets management, and an LLM pipeline that parses uploaded documents into structured, bookable jobs.
- Live with a pilot home service provider: 30 bookings across 8 users since launch. Own deployment, uptime, and incident response. — belashomemaintenance.com

QUANT SKILLS: Python (Pandas/NumPy/statsmodels/scipy), C++, Java, Factor Research & IC Analysis, Backtesting & Walk-Forward Validation, Portfolio Construction & Optimization, Statistical Arbitrage, Cointegration Analysis, Cross-Sectional Momentum, Transaction Cost Modeling, Regression Analysis, Statistical & Mathematical Modelling, Quantitative Trading, Discretionary Trading

INTERESTS: Poker, Chess, Squash, Skiing

LANGUAGES: English, Spanish